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Roll No

ME-701(A)-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Design of Heat Exchanger

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Explain with neat sketch different types of Heat Exchanger. Also write advantages and disadvantages of each.
2. Shell and tube type heat exchangers are the most common type of heat exchangers. Why? State its different applications in industries.
3. A shell and tube heat exchanger is to be designed for heating water from 25°C to 50°C with the help of steam condensing at atmospheric pressure. The water flows through the tubes (2.5cm ID and 2.9cm OD and 2m long) and the steam condenses on the outside. Calculate the number of tubes required if water flow rate is 500 kg/min and the individual coefficient of heat transfer on the steam and water sides are 8000 and 3000 W/m²-K. Neglect all other resistances.
4. Explain the working of Desert cooler with the help of neat sketch.

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5.
 - a) In a liquid to gas heat exchanger, it is best to put extended surfaces on the gas side. Why?
 - b) What are the causes of pressure drop in shell and tube heat exchanger.
6.
 - a) What are the causes of development of stress in a heat exchanger and how thermal stress can be minimized?
 - b) Show with neat sketch of temperature distribution for unmixed cross flow heat exchanger and explain it.
7. Explain different types of corrosion occur in a heat exchanger. Name different methods for testing a heat exchanger.
8. Write short notes (any two):
 - a) Heat pipe
 - b) Software in heat exchanger design
 - c) Cooling and dehumidifying coils
 - d) Design of air washer
